

Question 1

P1

A group of 7 friends decide to go to a movie.

Determine how many ways the friends can sit in a row if two of the friends refuse to sit next to each other.

Solution

$$7! - 6! \cdot 2! = 3600 \text{ ways}$$

$\frac{1}{2}$ mark for $7!$

1 mark for product of $6! \cdot 2!$

($\frac{1}{2}$ mark for $6!$, $\frac{1}{2}$ mark for $2!$)

$\frac{1}{2}$ mark for subtraction

2 marks

Question 2

R10

Gabrielle listens to her radio at a sound level of 80 dB. She attended a music concert that had a sound level of 115 dB. Determine how many times more intense the music concert was than the radio.

You may use the formula:

$$\beta = 10 \log \left(\frac{I}{I_0} \right)$$

where β is the intensity level of sound, measured in dB

I is the intensity of sound

I_0 is the standard minimum intensity that a person can hear

Solution

Radio:

$$80 = 10 \log \left(\frac{I}{I_0} \right)$$

$$8 = \log \left(\frac{I}{I_0} \right)$$

$$10^8 = \frac{I}{I_0} \quad \frac{1}{2} \text{ mark for exponential form}$$

$$10^8 I_0 = I$$

Music concert:

$$115 = 10 \log \left(\frac{I}{I_0} \right)$$

$$11.5 = \log \left(\frac{I}{I_0} \right)$$

$$10^{11.5} = \frac{I}{I_0} \quad \frac{1}{2} \text{ mark for exponential form}$$

$$10^{11.5} I_0 = I$$

$$\begin{aligned} \frac{\text{intensity of music concert}}{\text{intensity of radio}} &= \frac{10^{11.5} I_0}{10^8 I_0} \\ &= 10^{3.5} \\ &= 3162.27766 \\ &= 3162.278 \end{aligned}$$

1 mark for comparison

2 marks

Question 3

R10

Solve, algebraically.

$$2(7)^x = 3^{2x-3}$$

Solution

$$\log\left(2(7^x)\right) = \log 3^{2x-3}$$

½ mark for applying logarithms

$$\log 2 + x \log 7 = (2x - 3) \log 3$$

1 mark for product law

1 mark for power law

$$\log 2 + x \log 7 = 2x \log 3 - 3 \log 3$$

$$\log 2 + 3 \log 3 = 2x \log 3 - x \log 7$$

½ mark for collecting terms with x

$$\log 2 + 3 \log 3 = x(2 \log 3 - \log 7)$$

$$\frac{\log 2 + 3 \log 3}{2 \log 3 - \log 7} = x$$

½ mark for isolating x

$$15.872\ 483 = x$$

$$15.872 = x$$

½ mark for evaluating quotient of logarithms

4 marks

Solve for θ , algebraically, over the interval $[0, 2\pi]$.

$$\csc^2 \theta + 2 \csc \theta - 8 = 0$$

Solution

$$(\csc \theta + 4)(\csc \theta - 2) = 0$$

$$\csc \theta = -4 \quad \csc \theta = 2$$

$$\sin \theta = -\frac{1}{4} \quad \sin \theta = \frac{1}{2}$$

$$\theta_r = 0.252\ 680$$

$$\theta = 3.394 \quad \theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\theta = 6.031$$

or

$$\theta = 3.394 \quad \theta = 0.524$$

$$\theta = 6.031 \quad \theta = 2.618$$

1 mark for solving for $\csc \theta$

1 mark for reciprocal

2 marks for solving for θ ($\frac{1}{2}$ mark for each value)

4 marks

Question 5

P1

You have forgotten the code to unlock your cell phone. You know the code is made up of four numbers from 0 to 9.

Determine the number of possible codes, if repetition is allowed.

Solution

$$\underline{10} \cdot \underline{10} \cdot \underline{10} \cdot \underline{10} = 10\,000$$

1 mark

In the binomial expansion of $\left(\frac{7}{x^3} - 3x^7\right)^n$, the 5th term contains x^7 .

Determine the value of n .

Solution

$$x^7 = \left(\frac{1}{x^3}\right)^{n-4} (x^7)^4$$

1 mark for $k = 4$
½ mark for substitution

$$x^7 = (x^{-3})^{n-4} (x^7)^4$$

$$x^7 = x^{-3n+12+28}$$

$$7 = -3n + 40$$

$$-33 = -3n$$

$$11 = n$$

½ mark for solving for n

2 marks

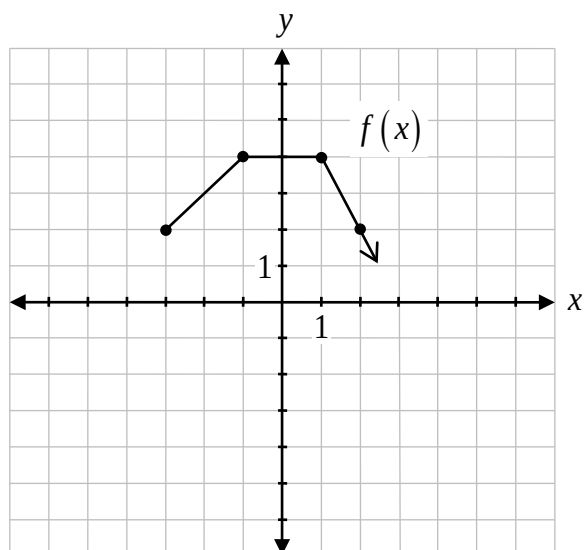
Given the domain of $f(x)$ is $\{-6, 1, 3, 4\}$ and the range of $f(x)$ is $\{-4, 7, 10, 15\}$, state the domain of $f^{-1}(x)$.

Solution $\{-4, 7, 10, 15\}$ **1 mark**

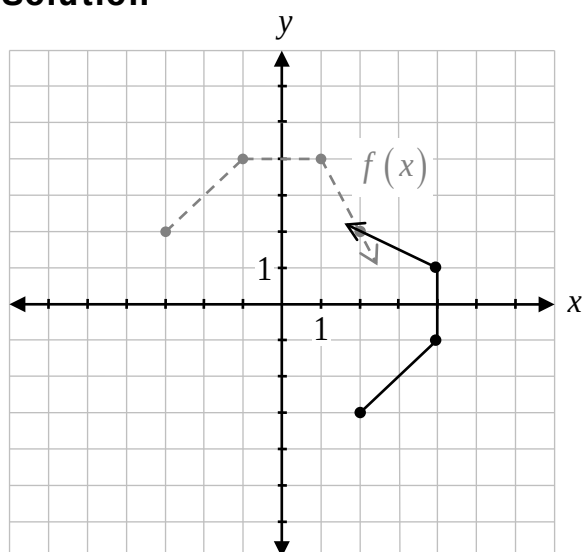
Question 8

R6

Given the graph of $y = f(x)$, sketch the graph of its inverse.



Solution



1 mark

Prove the following identity for all permissible values of θ .

$$\frac{1 + \cos \theta}{1 - \sin^2 \theta} = \sec \theta + \tan^2 \theta + 1$$

Solution

Method 1

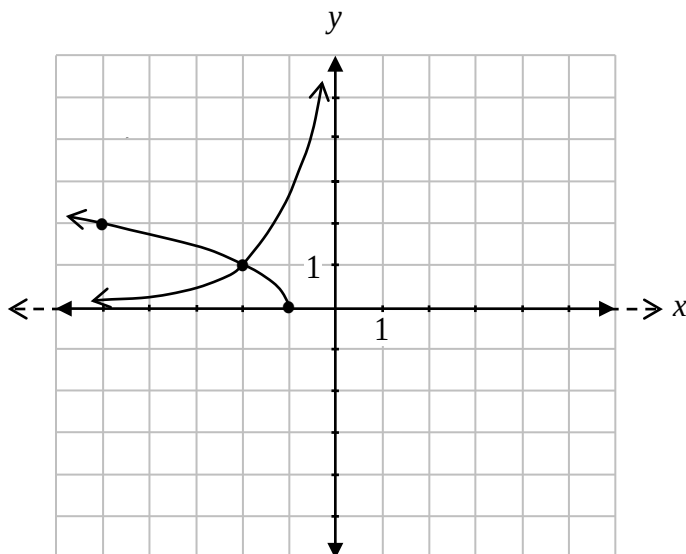
Left-Hand Side	Right-Hand Side
$\frac{1 + \cos \theta}{1 - \sin^2 \theta}$	$\sec \theta + \tan^2 \theta + 1$
$\frac{1 + \cos \theta}{\cos^2 \theta}$	1 mark for algebraic strategies
$\frac{1}{\cos^2 \theta} + \frac{1}{\cos \theta}$	1 mark for logical process to prove the identity
$\sec^2 \theta + \sec \theta$	1 mark for correct substitution of appropriate identities
$\tan^2 \theta + 1 + \sec \theta$	3 marks

Solution**Method 2**

Left-Hand Side	Right-Hand Side	
$\frac{1 + \cos \theta}{\cos^2 \theta}$	$\frac{1}{\cos \theta} + \sec^2 \theta$	
	$\frac{1}{\cos \theta} + \frac{1}{\cos^2 \theta}$	1 mark for algebraic strategies
	$\frac{\cos \theta + 1}{\cos^2 \theta}$	1 mark for logical process to prove the identity
		1 mark for correct substitution of appropriate identities

3 marks

Thomas used graphs to solve the equation $e^{x+2} = \sqrt{-(x+1)}$.



He incorrectly states the solution as $(-2, 1)$.

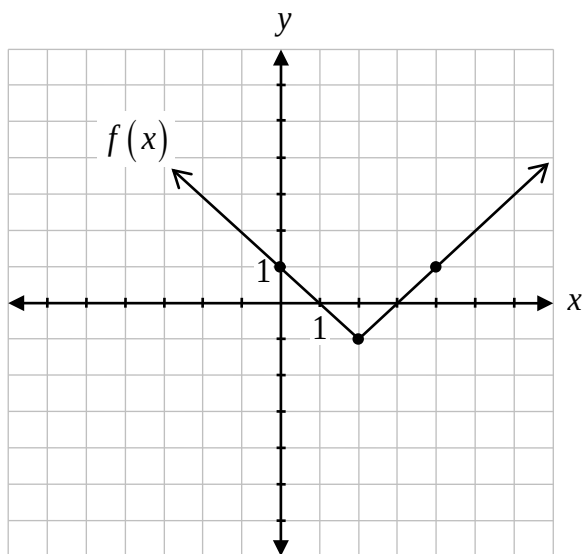
Describe how Thomas should have stated the solution.

Solution

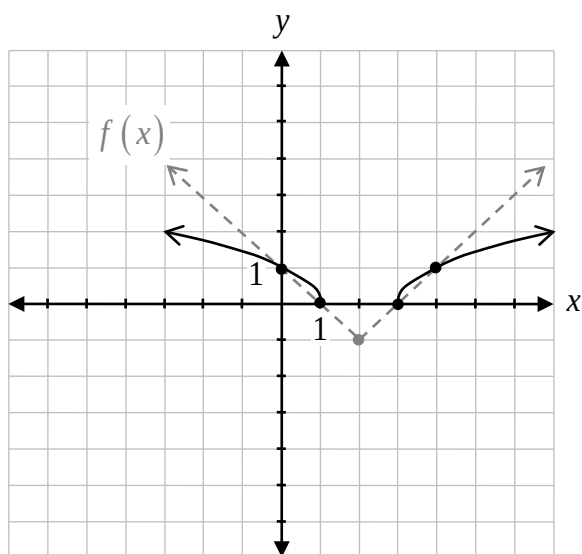
He stated his solution as a coordinate point; his solution should have only been the value of x .

1 mark

Given the graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.



Solution



1 mark for restricting domain
 $\frac{1}{2}$ mark for shape between both pairs of invariant points
 $\frac{1}{2}$ mark for shape above both pairs of invariant points

2 marks

When a polynomial, $P(x)$, is divided by $(x - 2)$ the resulting equation is

$$\frac{P(x)}{x - 2} = x^2 - x + 1 + \frac{3}{x - 2}.$$

- a) Explain why $x - 2$ is not a factor of $P(x)$.
- b) Determine the equation for the polynomial function $P(x)$.

Solution

- a) There is a remainder when $P(x)$ is divided by $x - 2$.

1 mark

b) $P(x) = \underline{(x - 2)(x^2 - x + 1) + 3}$

1 mark

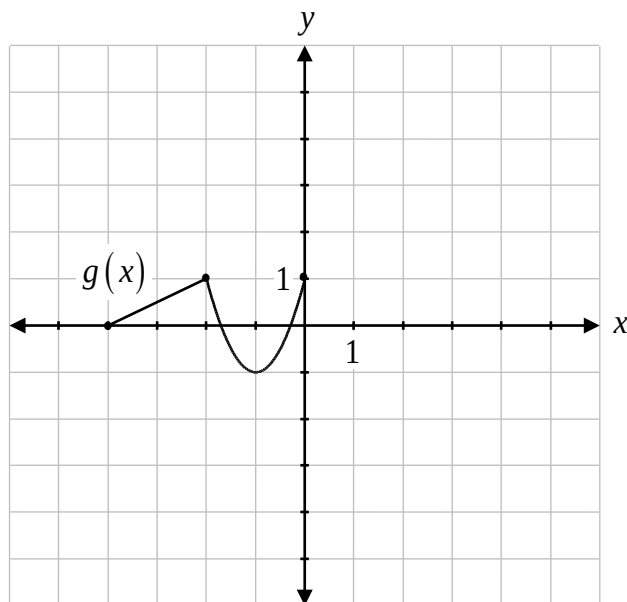
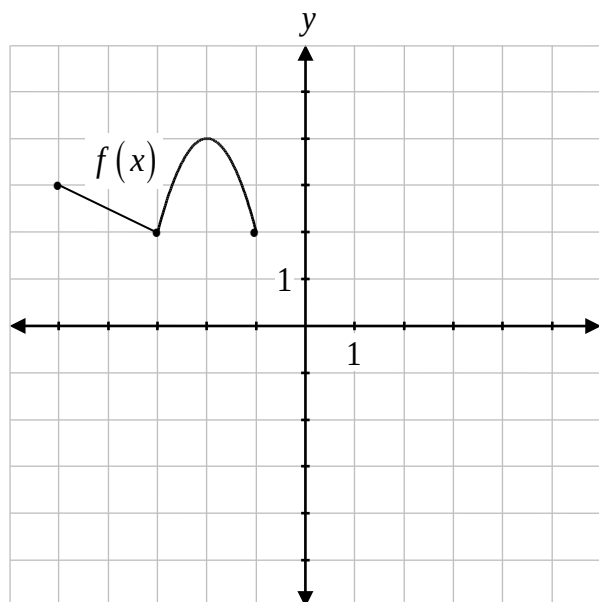
or

$$P(x) = \underline{x^3 - 3x^2 + 3x + 1}$$

Question 13

R2, R5

Determine the equation for $g(x)$ in terms of $f(x)$.



Solution

$$g(x) = -f(x-1) + 3$$

1 mark for vertical reflection
1 mark for horizontal translation
1 mark for vertical translation

3 marks

Explain why the binomial expansion of $(2x + y)^9$ does not have a middle term.

Solution

The expansion contains $n + 1$ terms. Since n equals 9, there are 10 terms, which would not allow for a middle term.

1 mark

Using the laws of logarithms, completely expand the expression $\log\left(\frac{5\sqrt{a}}{b^3}\right)$.

Solution

$$\log 5 + \frac{1}{2} \log a - 3 \log b$$

1 mark for product law
1 mark for power law (½ mark for each)
1 mark for quotient law

3 marks

Question 16

T1

Identify 10° in radians.

a) $\frac{1800}{\pi}$

b) $\frac{\pi}{1800}$

c) $\frac{18}{\pi}$

d) $\frac{\pi}{18}$

Question 17

R12

The polynomial function, $P(x) = a(x-1)^2(x+4)^2$, has a y-intercept of -8 .

Identify the value of a .

a) -2

b) $-\frac{1}{2}$

c) $\frac{1}{2}$

d) 2

Question 18

R7

Identify the value of $\log_4\left(\frac{1}{16}\right)$.

a) -2

b) $-\frac{1}{2}$

c) $\frac{1}{2}$

d) 2

Question 19

T1

Given the angle $\frac{25\pi}{7}$, identify the coterminal angle on the interval $[-2\pi, 0]$.

a) $\frac{18\pi}{7}$

b) $\frac{11\pi}{7}$

c) $-\frac{3\pi}{7}$

d) $-\frac{10\pi}{7}$

Question 20

P2

Identify which expression cannot be evaluated.

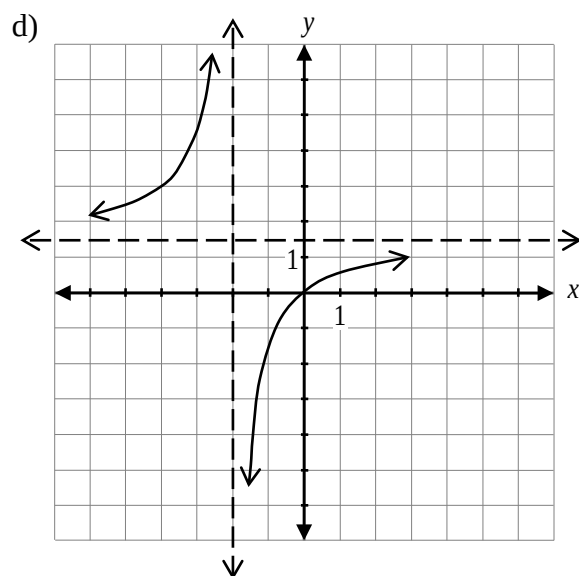
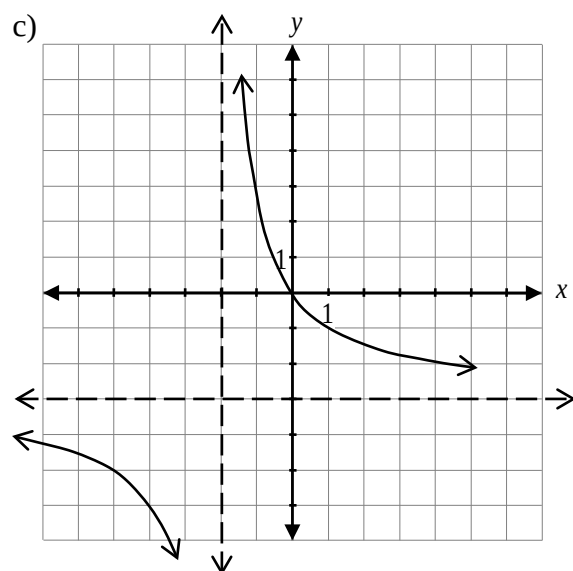
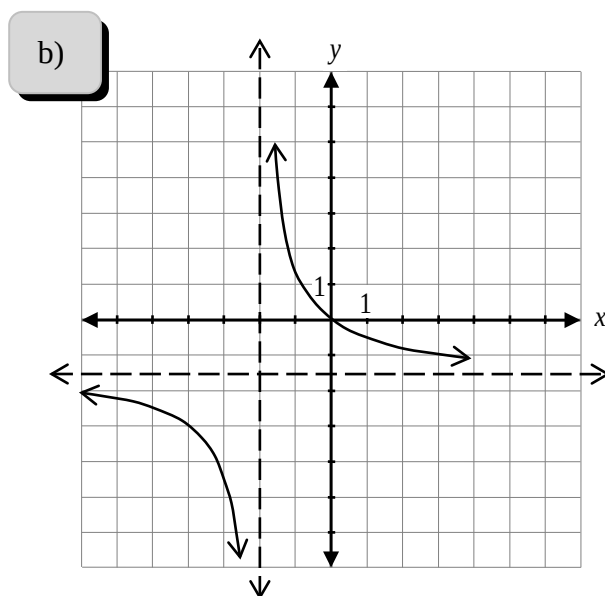
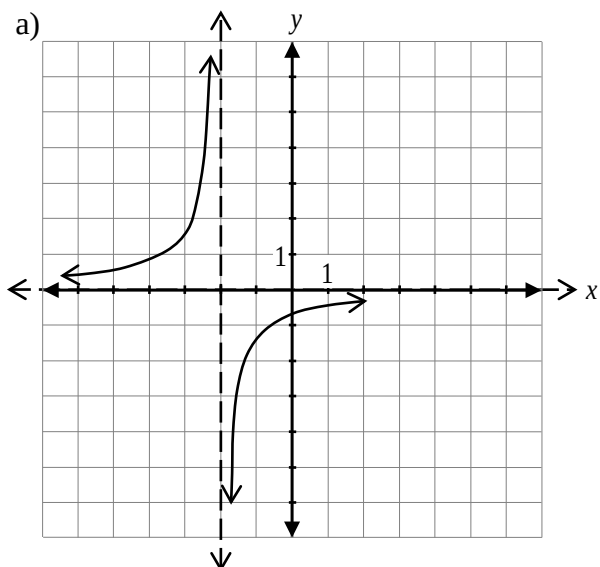
a) ${}_7P_0$

b) ${}_7P_6$

c) ${}_7P_7$

d) ${}_7P_8$

Identify the graph of $f(x) = \frac{-3x}{2x+4}$.



Question 22**R3**

Given a point $(-2, 0)$ on the graph of $y = f(x)$, identify the coordinates of the corresponding point on the graph of $y = 4f\left(\frac{1}{2}x\right)$.

a) $(-8, 0)$

b) $(-4, 0)$

c) $(-2, 0)$

d) $(-1, 0)$

Question 23**T6**

Identify the non-permissible value of θ for the expression $\frac{\cos \theta}{1 + \sin \theta}$.

a) $\frac{\pi}{2}$

b) π

c) $\frac{3\pi}{2}$

d) 2π

Question 24**R9**

Identify the function with an asymptote at $x = -3$.

a) $y = \log(x + 3)$

b) $y = \log x + 3$

c) $y = \log(x - 3)$

d) $y = \log x - 3$

Evaluate the following expression.

$$\tan\left(\frac{2\pi}{3}\right)\csc\left(\frac{-2\pi}{3}\right) + \cos(3\pi)$$

Solution

$$\begin{array}{l} (-\sqrt{3})\left(-\frac{2}{\sqrt{3}}\right) + (-1) \\ \frac{2-1}{1} \end{array}$$

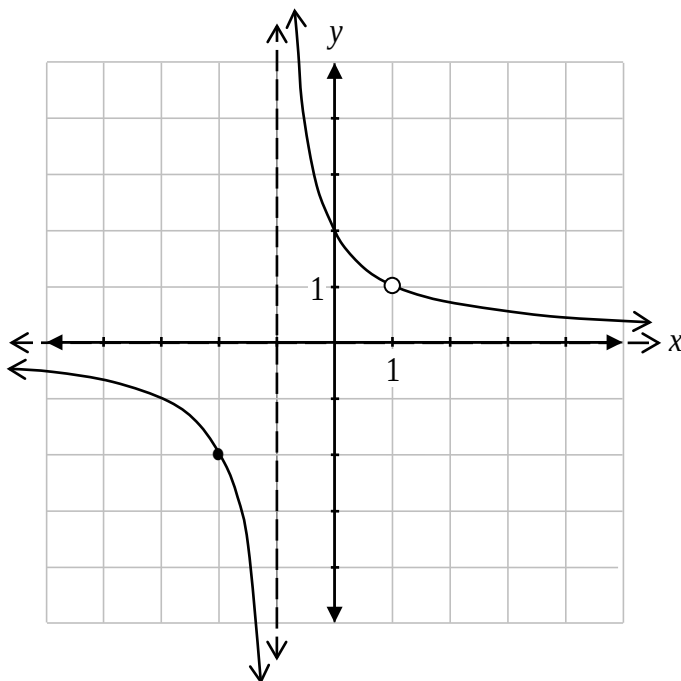
1 mark for $\tan\left(\frac{2\pi}{3}\right)$ (½ mark for quadrant, ½ mark for value)

1 mark for $\csc\left(-\frac{2\pi}{3}\right)$ (½ mark for quadrant, ½ mark for value)

1 mark for $\cos(3\pi)$

3 marks

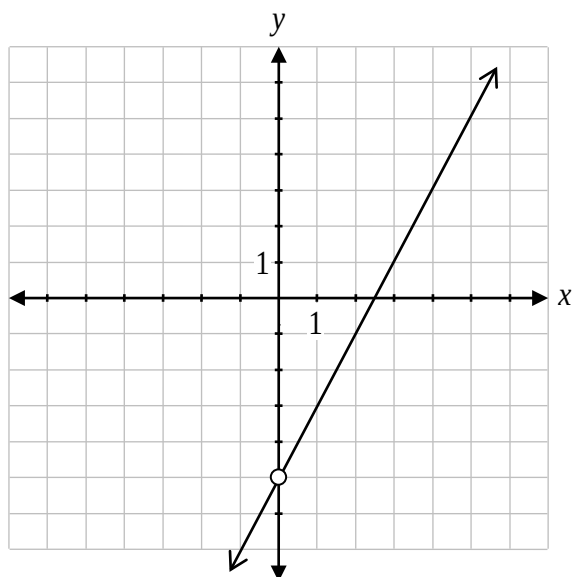
State the range of the graph below.

**Solution**

Range: $\{y \in \mathbb{R}, y \neq 0 \text{ and } y \neq 1\}$ 1 mark ($\frac{1}{2}$ mark for $y \neq 0$, $\frac{1}{2}$ mark for $y \neq 1$)

1 mark

Sketch the graph of the function $f(x) = \frac{2x^2 - 5x}{x}$.

Solution

1 mark for point of discontinuity (hole) at $(0, -5)$

($\frac{1}{2}$ mark for $x = 0$, $\frac{1}{2}$ mark for $y = -5$)

1 mark for shape of a linear function

2 marks

State a possible value of n if the polynomial function $P(x) = (x-1)^2(x+2)^n$ has a range of $[0, \infty)$.

Solution

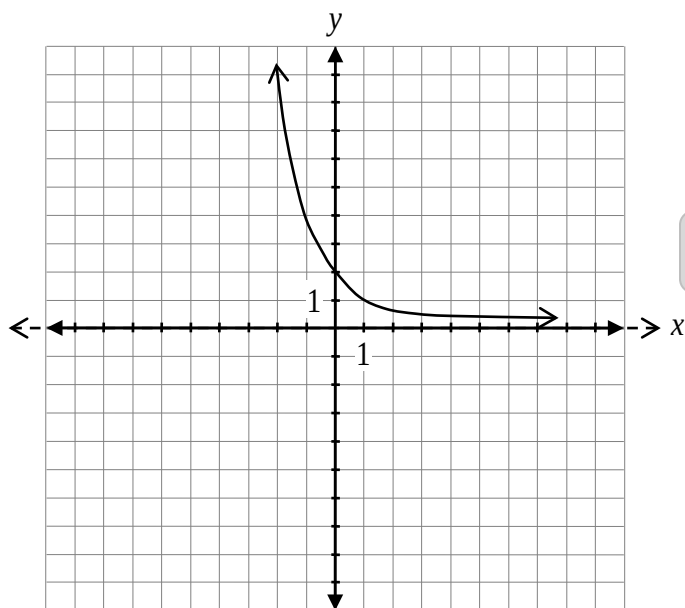
$$n = 2$$

1 mark

Note(s):

- Accept any even positive value of n , including zero.

Sketch the graph of $y = \left(\frac{1}{2}\right)^{x-1}$.

Solution

1 mark for decreasing exponential function
1 mark for horizontal translation

2 marks

Solve.

$$\log_x 27 = 3$$

Solution

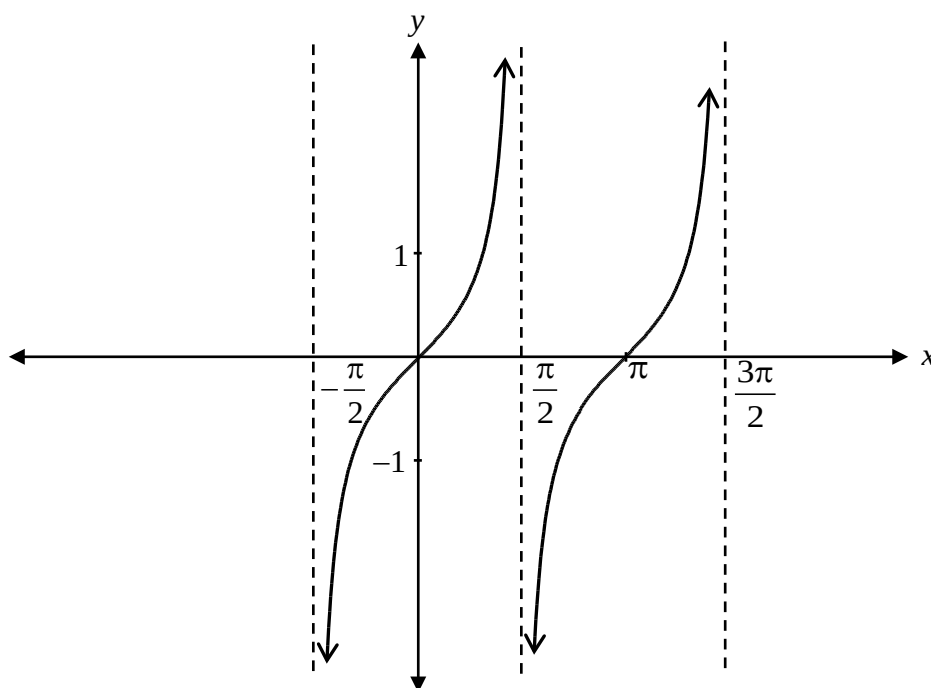
$$x^3 = 27$$

1 mark for exponential form

$$x = 3$$

1 mark

Sketch at least two periods of the graph $y = \tan x$.

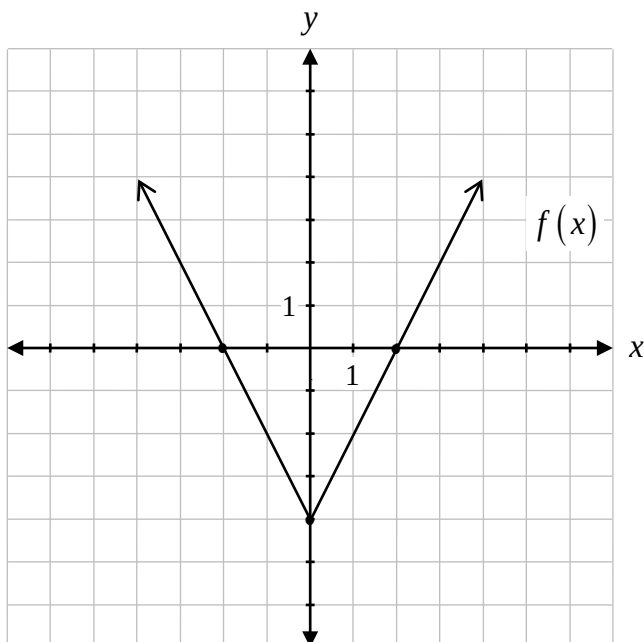
Solution

1 mark for increasing trigonometric function

1 mark for asymptotic behaviour approaching $x = \frac{\pi}{2} + k\pi$, $k \in \mathbb{Z}$

2 marks

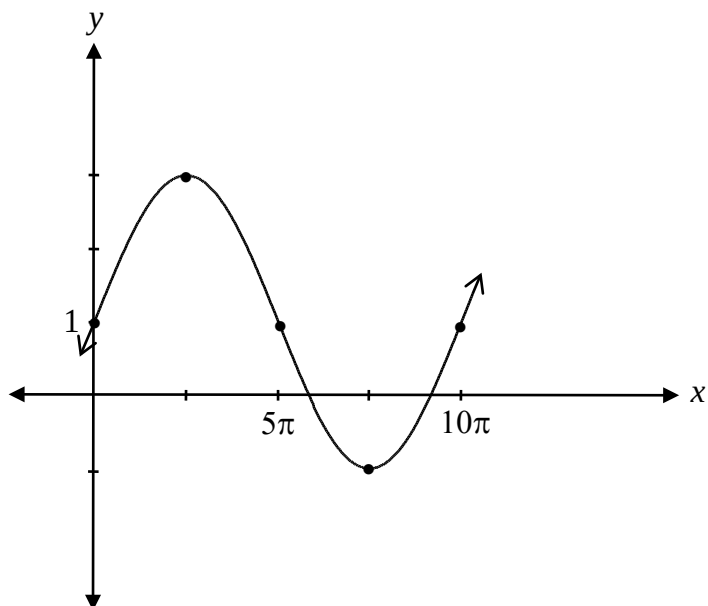
Given the graph of $f(x)$, state the domain of $\frac{1}{f(x)}$.

**Solution**

Domain: $\{x \in \mathbb{R}, x \neq \pm 2\}$ 1 mark (½ mark for $x \neq 2$, ½ mark for $x \neq -2$)

1 mark

Determine the values of A, B, and D of the sinusoidal function in the form $y = A \sin(Bx) + D$.

**Solution**

$$A = \underline{\quad 2 \quad}$$

1 mark for A

$$B = \underline{\quad \frac{1}{5} \quad}$$

1 mark for B

$$D = \underline{\quad 1 \quad}$$

1 mark for D

3 marks

Determine if the point $\left(-\frac{\sqrt{7}}{5}, \frac{2}{5}\right)$ is on the unit circle.

Justify your answer.

Solution

$$x^2 + y^2 = 1$$

$$\text{Left-hand side} = \left(-\frac{\sqrt{7}}{5}\right)^2 + \left(\frac{2}{5}\right)^2$$

$$= \frac{7}{25} + \frac{4}{25}$$

$$= \frac{11}{25}$$

$$\frac{11}{25} \neq 1$$

$\therefore$ not on the unit circle

1 mark for justification

1 mark

Solve, algebraically.

$$\frac{{}_n C_5}{{}_n C_4} = 6$$

Solution

$$\frac{\frac{n!}{(n-5)!5!}}{\frac{n!}{(n-4)!4!}} = 6$$

½ mark for substitution into equation

$$\frac{n!(n-4)!4!}{n!(n-5)!5!} = 6$$

$$\frac{\cancel{n!} (n-4) \cancel{(n-5)!} \cancel{4!}}{\cancel{n!} \cancel{(n-5)!} 5 \cdot \cancel{4!}} = 6$$

1 mark for factorial expansion

(½ mark for numerical factors; ½ mark for factors with variables)

1 mark for simplification of factorials

(½ mark for numerical factors; ½ mark for factors with variables)

$$\frac{n-4}{5} = 6$$

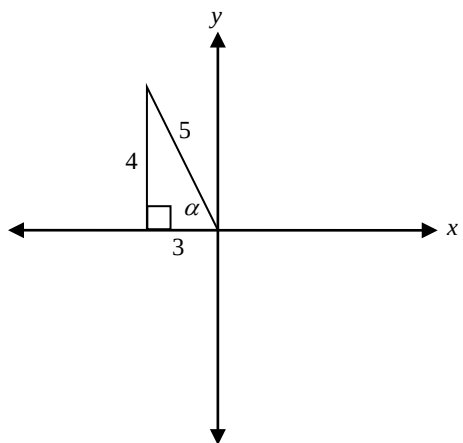
$$n-4 = 30$$

$$n = 34$$

½ mark for solving for n

3 marks

Given $\sin \alpha = \frac{4}{5}$, where α is in quadrant II, determine the exact value of $\sin 2\alpha$.

Solution

$$x^2 + y^2 = r^2$$

$$x^2 + 16 = 25$$

$$x^2 = 9$$

$$x = \pm 3$$

$$x = -3$$

$$\cos \alpha = -\frac{3}{5}$$

½ mark for value of x

½ mark for $\cos \alpha$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$= 2 \left(\frac{4}{5} \right) \left(-\frac{3}{5} \right)$$

$$= -\frac{24}{25}$$

1 mark for substitution into correct identity

2 marks

Note(s):

- Accept any of the following values for x : $x = \pm 3$; $x = -3$; or $x = 3$.

Given the functions $f(x) = x + 1$ and $g(x) = \sqrt{x}$,

a) determine the equation of $g(f(x))$.

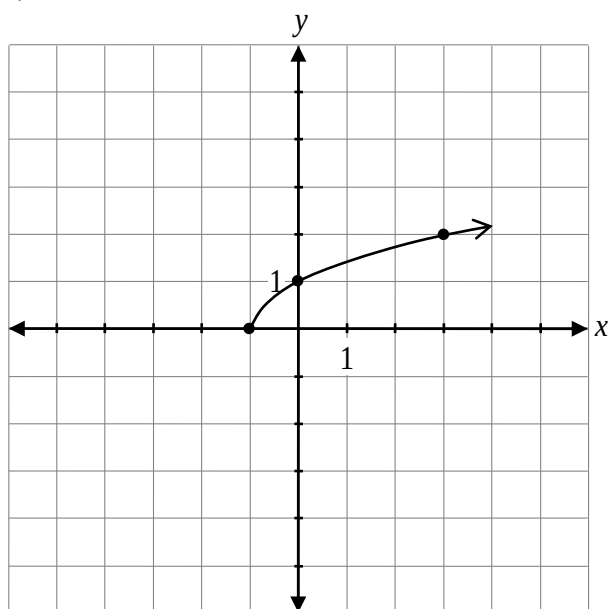
b) sketch the graph of $g(f(x))$.

Solution

a) $g(f(x)) = \sqrt{x+1}$ _____

1 mark

b)



1 mark for domain of $g(f(x))$

1 mark for shape consistent with $g(f(x))$

2 marks

Steve is asked to determine an equation with a larger period than the period of the graph of $y = \cos(2x)$.

Justify why Steve's answer of $y = \cos(6x)$ is incorrect.

Solution

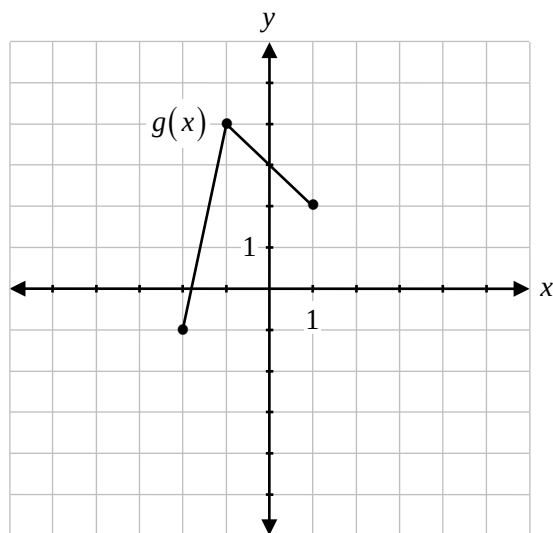
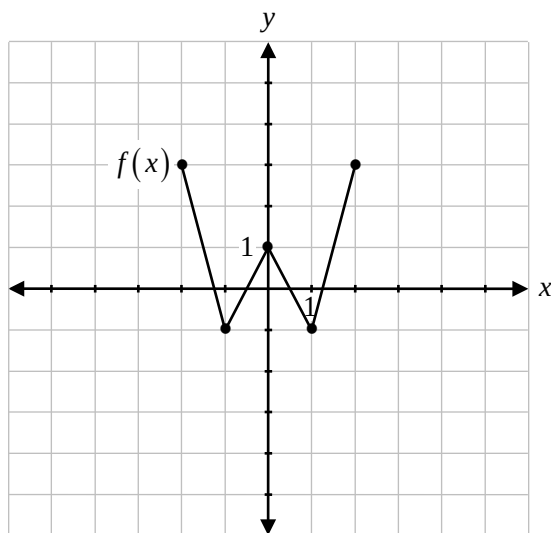
Steve's equation needs to have a value of $|b|$ less than 2.

1 mark

or

Steve's graph would have a period of $\frac{2\pi}{6} = \frac{\pi}{3}$, which is smaller than $\frac{2\pi}{2} = \pi$, the period of the given graph.

Given the graphs of $f(x)$ and $g(x)$,



a) determine the value of $(f \cdot g)(-1)$.

b) determine the value of $g(f(0))$.

Solution

$$\begin{aligned} \text{a) } (f \cdot g)(-1) &= (-1)(4) \\ &= -4 \end{aligned}$$

1 mark for value of $(f \cdot g)(-1)$

1 mark

$$\text{b) } f(0) = 1$$

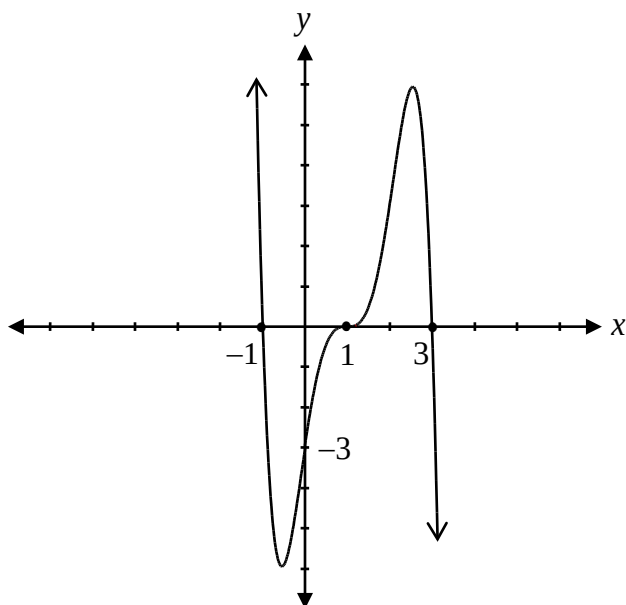
$\frac{1}{2}$ mark for $f(0)$

$$g(f(0)) = 2$$

$\frac{1}{2}$ mark for $g(f(0))$ consistent with $f(0)$ value

1 mark

Sketch the graph of $P(x) = -(x-1)^3(x-3)(x+1)$.

Solution

1 mark for x-intercepts

½ mark for y-intercept

1 mark for multiplicity (degree 3 at $x = 1$)

½ mark for end behaviour

3 marks

The point $(-\sqrt{3}, 1)$ is on the terminal arm of an angle θ , in standard position.

a) Determine $\tan \theta$.

b) Determine a possible value of θ , in radians.

Solution

a) $\tan \theta = -\frac{1}{\sqrt{3}}$

1 mark

b) $\theta = \frac{5\pi}{6}$

1 mark

Describe the transformation used to obtain the graph of $y = \log_5 x$ given the graph of $y = 5^x$.

Solution

The graph of $y = \log_5 x$ is obtained by reflecting the graph of $y = 5^x$ over the line $y = x$.

or

1 mark

The graph of $y = \log_5 x$ is the inverse of $y = 5^x$.

Solve $\sin \theta = -\frac{\sqrt{3}}{2}$, where $\theta \in \mathbb{R}$.

Solution

$$\theta = \frac{4\pi}{3}, \frac{5\pi}{3}$$

1 mark for values of θ ($\frac{1}{2}$ mark for each value)

$$\theta = \left\{ \begin{array}{l} \frac{4\pi}{3} + 2k\pi, k \in \mathbb{Z} \\ \frac{5\pi}{3} + 2k\pi, k \in \mathbb{Z} \end{array} \right\}$$

1 mark for general solution

2 marks

or

$$\theta = 240^\circ, 300^\circ$$

$$\theta = \left\{ \begin{array}{l} 240^\circ + 360^\circ k, k \in \mathbb{Z} \\ 300^\circ + 360^\circ k, k \in \mathbb{Z} \end{array} \right\}$$

Given that the point (a, b) is on the graph of $f(x)$, describe how you would determine the corresponding point on the graph of $y = \sqrt{f(x)}$.

Solution

The value of a stays the same, square root the value of b .

1 mark

Evaluate.

$$\cos\left(\frac{\pi}{20}\right)\cos\left(\frac{\pi}{5}\right) - \sin\left(\frac{\pi}{20}\right)\sin\left(\frac{\pi}{5}\right)$$

Solution

$$\cos\left(\frac{\pi}{20} + \frac{\pi}{5}\right)$$

½ mark for substitution of an appropriate identity

$$\cos\left(\frac{\pi}{20} + \frac{4\pi}{20}\right)$$

$$\cos\left(\frac{5\pi}{20}\right)$$

$$\cos\left(\frac{\pi}{4}\right)$$

$$\frac{\sqrt{2}}{2}$$

½ mark for exact value

1 mark

Describe the transformations used to obtain the graph of the function $y = f(-x + 6) - 8$ from the graph of $y = f(x)$.

Solution

Reflect the graph of $y = f(x)$ over the y -axis and then translate 6 units right and 8 units down.

1 mark for horizontal reflection
1 mark for horizontal translation
1 mark for vertical translation

3 marks

Note(s):

- Deduct 1 mark if correct transformations are given in the wrong order.

State the equations of all the asymptotes of the function, $y = \frac{1}{3x+1}$.

Solution

$$y = 0$$

1 mark for horizontal asymptote

$$x = -\frac{1}{3}$$

1 mark for vertical asymptote

2 marks

Determine the zeros of the polynomial function $P(x) = 2x^3 + 5x^2 - 4x - 3$.

Solution

$$P(1) = 2(1)^3 + 5(1)^2 - 4(1) - 3 \quad 1 \text{ mark for identifying one possible value of } x$$

$$P(1) = 0$$

$(x - 1)$ is a factor

$$\begin{array}{r|rrrr} 1 & 2 & 5 & -4 & -3 \\ & \downarrow & & & \\ & 2 & 7 & 3 & 0 \end{array}$$

1 mark for synthetic division (or for any equivalent strategy)

$$P(x) = (x - 1)(2x^2 + 7x + 3) \quad \frac{1}{2} \text{ mark for consistent factors}$$

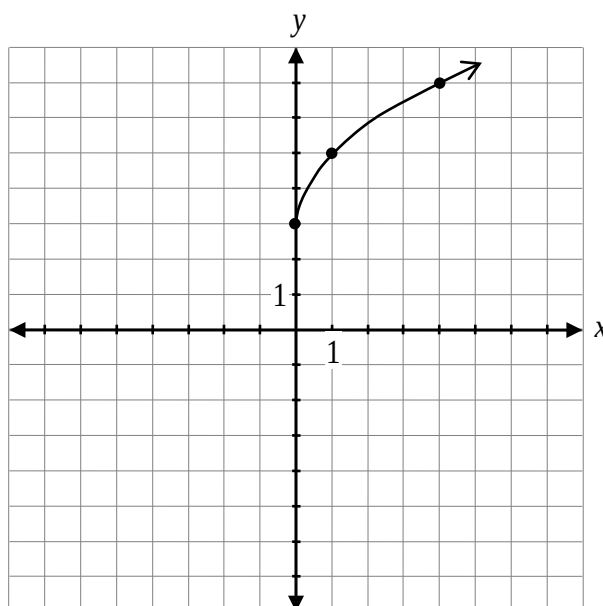
$$0 = (x - 1)(2x + 1)(x + 3)$$

$$x = 1 \quad x = -\frac{1}{2} \quad x = -3$$

$\frac{1}{2}$ mark for all zeros

3 marks

Determine the equation of the radical function represented by the graph.

**Solution**

$$y = 2\sqrt{x} + 3$$

1 mark for vertical stretch
1 mark for vertical translation

or

$$y = \sqrt{4x} + 3$$

1 mark for horizontal stretch
1 mark for vertical translation

2 marks